Understanding Community-Level Predictors of Obesity Categories

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Problem Definition and Overall Goal

Our project examines obesity levels, categorized into high, medium, and low, using sociodemographic characteristics as predictors. The goal is to understand which factors are meaningfully associated with being in each obesity group and to study broad patterns rather than predict exact percentages. We used classification because grouping tracts into three obesity levels is easier to interpret for inference and allows us to compare how obesity patterns differ across communities. We considered modeling obesity percentage directly, but that level of precision is not needed for our goal of identifying general associations across groups, so classification is a better fit for our study. We focused on obesity because it is closely linked to long-term health outcomes and overall well-being, and examining variation across tracts helps reveal how demographic and socioeconomic conditions relate to differences in obesity prevalence. Since our work centers on inference, the model is used only to describe associations between predictors and obesity categories. Policy makers may eventually read inference results to design interventions, but the model itself is not prioritizing or ranking tracts or making recommendations.

Data and Choice of Features

Our socioeconomic and demographic predictors come from the American Community Survey (ACS) 2015 5-Year Estimates. We use variables such as median household income, poverty rate, unemployment rate, educational attainment, median age, and racial or ethnic composition. These features were chosen because demographic and socioeconomic characteristics are stable community attributes that help us understand patterns related to obesity without directly incorporating health behaviors that could introduce reverse causality or data leakage. They also reduce collinearity and allow for more interpretable associations in an inference setting.

Our outcome variable, obesity prevalence, was obtained from the CDC 500 Cities Project. To align with our classification framework, we grouped obesity percentages into three levels: low, medium, and high, based on 33 percent quantile cutoffs. Categorizing the outcome makes patterns easier to interpret and better suited for describing how different community characteristics relate to higher or lower obesity levels.

Predictive Model Setup

For both of our model classes, we wanted to ensure that there was no data leakage. To do so, we made sure to split by county since nearby tracts may not be independent, and random splits would cause spatial leakage. This ensures true generalization to new geographic cases. In particular, we split the data into a 60/20/20 train/test/validation split, ensuring counties did not cross splits. We also used GroupKFold by county as cross validation on the train set only. An additional check we took into account was ensuring the target distribution of obesity percentage is similar across splits

Our first model approach is a k-Nearest Neighbors classifier. We selected this method because it can effectively capture local patterns in tract-level health data and accommodate non-linear relationships between socioeconomic characteristics and obesity prevalence. The key strengths of a kNN classifier model include its non-parametric nature, flexibility, and reliance on similarity-based classification. However, kNN can be computationally expensive and is highly sensitive to noise and unscaled variables. In the context of our overall goal, this limitation is less concerning because the model will be run infrequently, approximately once every few years, to update tract-level classifications as new obesity data become available. The primary assumptions underlying this model are that features are properly scaled and relevant, and that tracts with similar socioeconomic profiles are likely to exhibit similar obesity outcomes.

For our second model, we chose to do a Logistic Regression. Specifically, a Multinomial Logistic Regression for its ability to model obesity category outcomes efficiently. Providing highly interpretable results. This allows us to examine how socioeconomic features shift the likelihood of belonging to each obesity prevalence category. The model models categorical outcomes well through probability estimates. This made it ideal for us to understand how socioeconomic factors relate to obesity categories (high, medium, and low). One of the main strengths of the model is its transparency. The coefficients can be translated into odds ratios, enabling interpretable results that can help one infer how variables such as income, education, or access to healthcare influence categorical obesity outcomes on a tract level. However, the model is limited in its ability to model complex relationships or non-linear relationships, leaving room for not much flexibility, but in our case, the model aligns with our goal. The primary assumptions in this approach are a linear relationship between predictors and the obesity category, and it also assumes independent observations and no perfect multicollinearity among predictors.

Evaluation and Results

Figure 1:

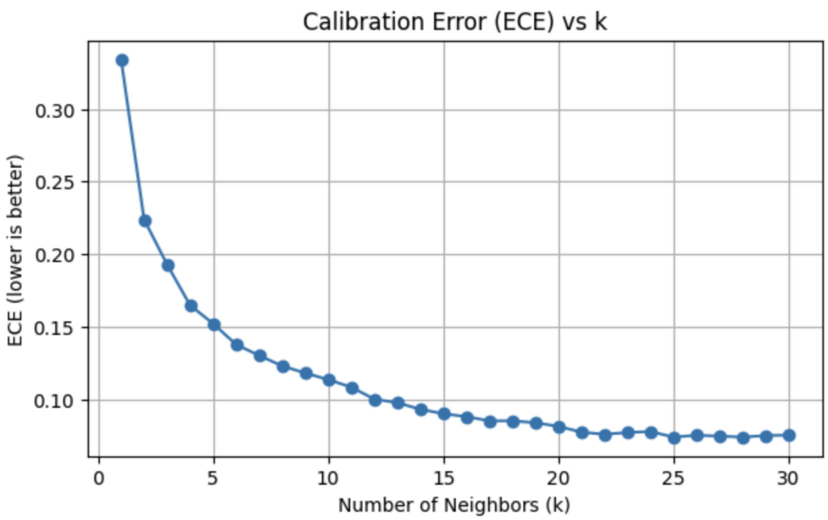

Figure 2:

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Final KNN model (k=28) evaluation:
Accuracy: 0.7324
Log Loss: 0.7227
ECE: 0.0739

Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.80      0.81      2568
           1       0.63      0.59      0.61      2308
           2       0.74      0.81      0.77      1862

    accuracy                           0.73      6738
   macro avg       0.73      0.73      0.73      6738
weighted avg       0.73      0.73      0.73      6738


Confusion Matrix:
[[2066  462   40]
 [ 465 1362  481]
 [  10  345 1507]]
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To determine the optimal number of neighbors for our k-Nearest Neighbors classifier model, we prioritized probability calibration rather than traditional methods like AUC. Because our goal was to capture broad patterns in obesity level distributions rather than optimize pure classification performance, we evaluated each value using expected calibration errors. This quantifies how closely predicted probabilities align with observed class frequencies. We plotted ECE values v k to determine the best k value (Figure 1). As expected, very small k values produced noisy and poorly calibrated probability estimates, while very large k values risk overwhelming meaningful local structure. The calibration curve showed a consistent decline in ECE as k increased; however, it levels off in the upper 20s, so we selected k = 28. This offered strong calibration while maintaining reasonable accuracy and class performance. The final evaluation metrics can be seen in Figure 2. Since kNN is nonparametric and has no training phases, this model does not optimize a loss function during training.

For evaluating our multinomial logistic regression, we also focused on probability calibration, examining how well the model's predicted probabilities aligned with observed class frequencies. We computed ECE across a range of regularization strengths using the inverse parameter C. With smaller C values having stronger regularization and larger C values, the opposite effect occurs. We selected a final C value of 0.01 because it provided the best calibration among the other tested values while maintaining a solid classification performance. The model achieved an accuracy of 0.73, with balanced performances across Low and medium classes, but had a lower recall for the high category. While the model's linear structure captures broad differences, the model struggled to capture complex patterns, specifically in High obesity prevalence. Even when tuning was undergone, the model remained poorly calibrated after adjustments were made, causing a limitation within the logistic regression model, given our data. Suggesting the model is not flexible enough to capture the relationships present in the tract-level obesity data.

In the context of modeling census tract obesity levels, we determined that the kNN model was ultimately the stronger classifier than the multinomial logistic regression. This model offered better probability calibration and greater flexibility for capturing complex, nonlinear community patterns. The nonparametric nature of the kNN model also allowed it to adapt to local variations in socioeconomic and demographic environments without imposing a rigid structural form. In contrast, the linear decision boundaries of the logistic regression constrained its ability to model the intricate interactions of tract-level data. This led to persistent miscalibration even after tuning the regularization strength. Lastly, though kNN takes longer to run, this is a model we would only run once every few years, which means that we can prioritize the accuracy and strength of this model over its running time.

Explanation and Interpretation of Results

The interpretation of results is a crucial step in our project because it allows us to move beyond mere prediction and understand the underlying patterns linking sociodemographic characteristics to obesity levels. Rather than focusing on precise obesity percentages, our goal is to identify which factors are meaningfully associated with being in the low, medium, or high obesity categories and to understand how changes in these factors relate to shifts between categories. The interpretation will be done using our kNN model, as it could capture complex, nonlinear patterns in the tracts.

SHAP is used to quantify the contribution of each feature to the predicted class for individual observations. A feature importance bar chart derived from SHAP values highlights which sociodemographic factors are most influential in distinguishing between obesity groups, offering clear insights into the patterns captured by the model.

Figure 3:

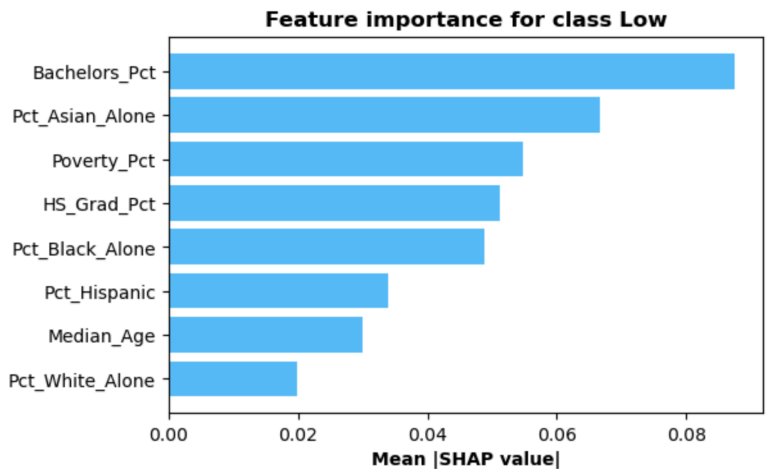


Figure 4:

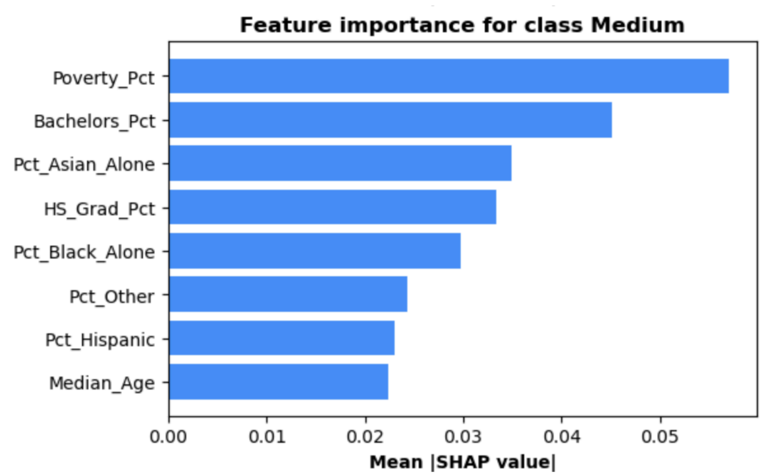
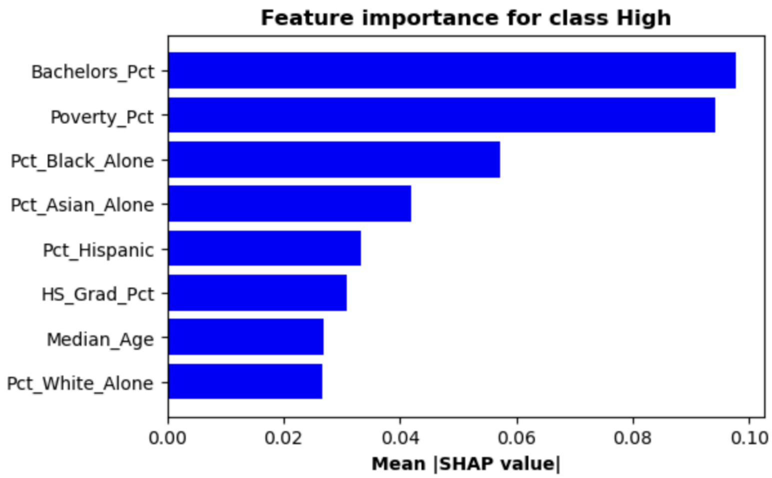


Figure 5:



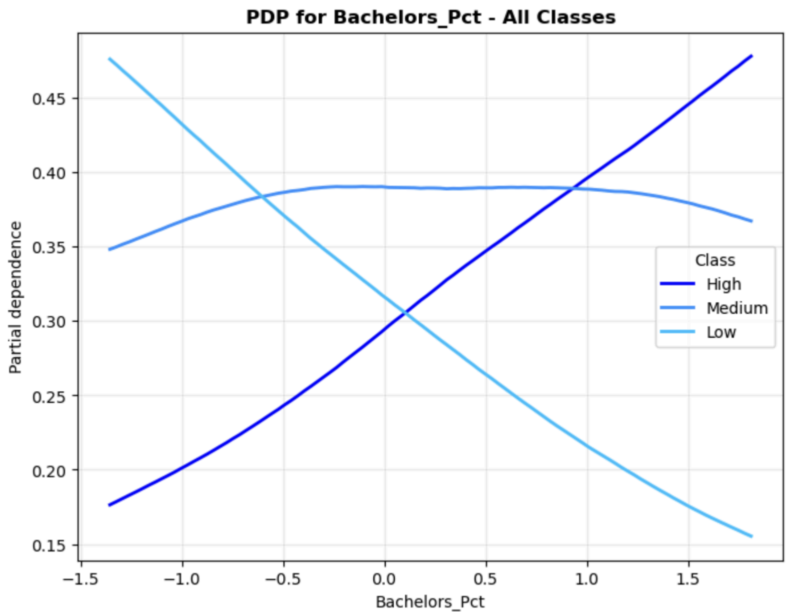
The SHAP feature importance plots (Figures 3–5) highlight which sociodemographic characteristics most strongly influence obesity classification. Across the low, medium, and high obesity categories, the percentage with the highest degree attainment being a bachelor's degree, poverty rate, and percent of Asian alone emerge as the most influential features. These plots show which variables the KNN model relies on most when distinguishing between the obesity categories, providing a clear view of the factors contributing to the model's predictions.

To better understand how individual features relate to obesity classification, partial dependence plots are used. These plots illustrate the effect of varying a single predictor on the predicted probabilities of each obesity category while averaging out the influence of other features. By focusing on features identified as most important, such as poverty rate and the percentage with a bachelor's degree, these plots provide insight into how changes in these variables are associated with shifts in the likelihood of being classified into the low, medium, or high obesity groups. The x-axis values for each feature are shown in their scaled form, reflecting how many standard deviations each observation is from the mean, which aligns with the preprocessing applied before modeling.

Figure 6:

The partial dependence plot for poverty percentage shows distinct patterns across the obesity categories. As the poverty percentage increases, the predicted probability of being in the low obesity category rises, while the probabilities for the medium and high categories decrease in parallel, with the medium category consistently higher than the high category. This suggests that higher poverty values are associated with a greater likelihood of classification in the low obesity group and a lower likelihood in the medium and high groups, with the medium category generally remaining more probable than the high category across the range of the feature.

Figure 7:



The partial dependence plot for the percentage of adults with a bachelor's degree shows contrasting trends across the obesity categories. As the percentage of adults with a bachelor's degree increases, the predicted probability of being in the low obesity category decreases, the probability for the medium category remains relatively constant, and the probability for the high obesity category increases. This indicates that higher values of this feature are associated with a higher likelihood of classification in the high obesity group and a lower likelihood in the low obesity group, while the medium group remains largely unaffected.

Overall, the SHAP feature importance analysis and partial dependence plots provide complementary insights into the factors associated with obesity classification. SHAP values identify which sociodemographic features the KNN model relies on most, highlighting poverty rate, percentage with a bachelor's degree, and racial composition as key predictors. Partial dependence plots then illustrate how variations in these features relate to changes in predicted probabilities across the low, medium, and high obesity categories. Together, these analyses reveal the patterns captured by the model and clarify the relative influence and behavior of the most important features, offering a clear understanding of how different sociodemographic characteristics contribute to distinguishing between obesity levels.

Discussion

Prior research consistently shows that obesity is strongly linked to socioeconomic and demographic conditions. National studies from the CDC (Centers for Disease Control and Prevention, 2023) and public-health literature (Ogden et al., 2015; Wang & Beydoun, 2007) find that lower income, higher poverty, higher unemployment, and lower educational attainment are associated with higher obesity prevalence. Demographic patterns are also well documented: counties with larger Black or Hispanic populations tend to experience higher obesity rates, while counties with more White or Asian residents generally show lower rates, reflecting long-standing structural and environmental inequities (Hales et al., 2020; Wang et al., 2008).

Our results are broadly consistent with these findings. When we classify counties into high, medium, and low obesity groups, we observe that economically disadvantaged counties and those with lower education levels are more likely to fall into the high-obesity category. Racial and ethnic composition also shows associations similar to those reported in prior work.

The contribution of our analysis is that we examine all sociodemographic factors jointly rather than one at a time. This multivariable perspective allows us to see which predictors remain meaningful after accounting for the others and provides a clearer picture of the relative importance of economic versus demographic characteristics. By structuring obesity as a three-level outcome, we highlight comparative differences across counties and reinforce known patterns while offering a more integrated view of how these factors collectively relate to obesity levels.

References

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